Problem 16.1.1

Expand the function in a Laurent series that converges for $0 < |z| < R$ and determine the precise region of convergence.

$$\frac{\cos z}{z^4}$$

Solution

The Taylor series for $\cos z$ is,

$$\cos z = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n},$$

Dividing by z^4,

$$\boxed{\frac{\cos z}{z^4} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n-4}}$$

The Taylor series for $\cos z$ converges with $R = \infty$, so the Laurent series converges for every $z \neq 0$, or $0 < |z| < \infty$.

Problem 16.1.3

Expand the function in a Laurent series that converges for $0 < |z| < R$ and determine the precise region of convergence.

$$\frac{\exp z^2}{z^3}$$

Solution

The Taylor series for $\exp z^2$ is,

$$\exp z^2 = \sum_{n=0}^{\infty} \frac{z^{2n}}{n!}$$

Dividing by z^3,

$$\boxed{\frac{\exp z^2}{z^3} = \sum_{n=0}^{\infty} \frac{z^{2n-3}}{n!}}$$

The Taylor series for $\exp z^2$ converges with $R = \infty$, so the Laurent series converges for every $z \neq 0$, or $0 < |z| < \infty$.

Problem 16.1.5

Expand the function in a Laurent series that converges for $0 < |z| < R$ and determine the precise region of convergence.

$$\frac{1}{z^2 - z^3}$$

Solution

$$\frac{1}{z^2 - z^3} = \frac{1}{z^2}\frac{1}{1-z}$$

The Taylor series for the geometric series $\frac{1}{1-z}$ is,

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$$

Dividing by z^2,

$$\boxed{\frac{1}{z^2 - z^3} = \sum_{n=0}^{\infty} z^{n-2}}$$

The Taylor series for $\frac{1}{1-z}$ converges on $|z| < 1$, so the Laurent series converges on $0 < |z| < 1$.

Problem 16.1.8

Expand the function in a Laurent series that converges for $0 < |z| < R$ and determine the precise region of convergence.

$$\frac{e^z}{z^2 - z^3}$$

Solution

The Taylor series for e^z is,

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

Multiplying by the series from **16.1.5**,

$$\frac{e^z}{z^2 - z^3} = \sum_{n=0}^{\infty} \frac{z^{2n-2}}{n!}$$

The Taylor series for e^z converges everywhere. As in **16.1.5**, the Laurent series converges on $0 < |z| < 1$.

Problem 16.1.13

Find the Laurent series that converges for $0 < |z-i| < R$ and determine the precise region of convergence.

$$\frac{1}{z^3 (z-i)^2}$$

Solution

Substituting $w = z - i$,

$$\begin{aligned}
\frac{1}{z^3 (z-i)^2} &= \frac{1}{w^2} \cdot \frac{1}{(i+w)^3} \\
&= \frac{1}{w^2} \cdot \frac{1}{i^3 \left(1 + \frac{w}{i}\right)^3} \\
&= \frac{1}{w^2} \cdot \frac{1}{-i (1 - iw)^3} \\
&= \frac{i}{w^2} \cdot \frac{1}{(1 - iw)^3}
\end{aligned}$$

The Taylor series of $\frac{1}{(1-iw)^3}$ is

$$\begin{aligned}
(1+z)^{-m} &= \sum_{n=0}^{\infty} \binom{-m}{n} z^n \\
(1-iw)^{-3} &= \sum_{n=0}^{\infty} \binom{-3}{n} (-iw)^n
\end{aligned}$$

Multiplying by $\frac{i}{w^2}$,

$$\frac{i}{w^2 (1-iw)^3} = \sum_{n=0}^{\infty} \binom{-3}{n} - i^{n+1} w^{n-2}$$

Substituting $z = w + i$,

$$\boxed{\frac{1}{z^3 (z-i)^2} = \sum_{n=0}^{\infty} \binom{-3}{n} - i^{n+1} z - i^{n-2}}$$

The Laurent series converges on $0 < |z-i| < 1$.

Problem 16.1.22

Find all Taylor and Laurent series with center $z_0 = i$. Determine the precise regions of convergence.

$$\frac{1}{z^2}$$

Solution

$\frac{1}{z^2}$ is singular at $z = 0$, therefore a Taylor series can be defined on $|z - i| < 1$ and a Laurent series can be defined on $|z - i| > 1$

Taylor Series

Substituting $w = z - i$,

$$\frac{1}{z^2} = \frac{1}{(i+w)^2} = \frac{1}{i^2(1-iw)^2} = \frac{-1}{(1-iw)^2}$$

The Taylors series of $-(1 - iw)^{-2}$ is

$$\begin{aligned} (1+z)^{-m} &= \sum_{n=0}^{\infty} \binom{-m}{n} z^n \\ -(1-iw)^{-2} &= -\sum_{n=0}^{\infty} \binom{-2}{n} (-iw)^n \end{aligned}$$

Substituting $z = w + i$,

$$\boxed{\frac{1}{z^2} = -\sum_{n=0}^{\infty} \binom{-2}{n} (-i)^n (z-i)^n}$$

Laurent Series

Substituting $w = z - i$ and rearranging for the Laurent series,

$$\frac{1}{z^2} = \frac{1}{(w+i)^2} = \frac{1}{w^2(1+\frac{i}{w})^2} = \frac{1}{w^2} \cdot \frac{1}{(1+\frac{i}{w})^2}$$

The Taylor series of $(1 + \frac{i}{w})^{-2}$

$$\begin{aligned} (1+z)^{-m} &= \sum_{n=0}^{\infty} \binom{-m}{n} z^n \\ (1-iw)^{-2} &= \sum_{n=0}^{\infty} \binom{-2}{n} \left(\frac{i}{w}\right)^n \end{aligned}$$

Dividing by w^2,

$$\frac{1}{w^2(w+i)^2} = \sum_{n=0}^{\infty} \binom{-2}{n} \frac{i^n}{w^{n+2}}$$

Substituting $z = w + i$,

$$\boxed{\frac{1}{z^2}\sum_{n=0}^{\infty}\binom{-2}{n}\frac{i^n}{(z-i)^{n+2}}}$$

Problem 16.1.23

Find all Taylor and Laurent series with center $z_0 = 0$. Determine the precise regions of convergence.

$$\frac{z^8}{1-z^4}$$

Solution

$\frac{z^8}{1-z^4}$ is singular at $z = \pm 1, \pm i$, therefore a Taylor series can be defined on $|z| < 1$ and a Laurent series can be defined on $|z| > 1$

Taylor Series

$$\frac{1}{1-z^4} = \sum_{n=0}^{\infty} z^{4n}$$

$$\boxed{\frac{z^8}{1-z^4} = \sum_{n=0}^{\infty} z^{4n+8}}$$

Laurent Series

$$\frac{1}{1-z^4} = \frac{-z^4}{1-z^{-4}} = -z^4 \sum_{n=0}^{\infty} z^{-4n}$$

$$\boxed{\frac{z^8}{1-z^4} = \sum_{n=0}^{\infty} z^{4-4n}}$$

Problem 16.2.1

Determine the location and order of the zeros.

$$\sin^4 \frac{1}{2}z$$

Solution

$$\begin{aligned}
&\sin^4 \frac{1}{2}z = 0 \\
&\implies \sin \frac{1}{2}z = 0 \\
&\implies \frac{1}{2}z = n\pi \\
&\implies \boxed{z = 2n\pi, \quad n \in \mathbb{Z} \text{ with order } 4}
\end{aligned}$$

Problem 16.2.3

Determine the location and order of the zeros.

$$(z + 81i)^4$$

Solution

$$\begin{aligned}
&(z + 81i)^4 = 0 \\
&\implies z + 81i = 0 \\
&\implies \boxed{z = -81i \text{ with order } 4}
\end{aligned}$$

Problem 16.2.5

Determine the location and order of the zeros.

$$z^{-2}\sin^2 \pi z$$

Solution

$$\begin{aligned} & z^{-2}\sin^2 \pi z = 0 \\ \implies & \sin^2 \pi z = 0 \\ \implies & z = n, \quad n \in \mathbb{Z} \end{aligned}$$

$$\sin^2 \pi z = \left(\sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} \right)^2 = (\pi z)^2 + \cdots \text{ (higher order powers of z)}$$

$$z^{-2}\sin^2 \pi z = \pi^2 + \cdots \text{ (higher order powers of z)}$$

Therefore $z = 0$ is a removable singularity and the zeros are

$$\boxed{z = \pm n, \quad n \in \mathbb{Z}^+ \text{ with order 2}}$$

Problem 16.2.7

Determine the location and order of the zeros.

$$z^4 + (1-8i)z^2 - 8i$$

Solution

$$\begin{aligned} z^4 + (1-8i)z^2 - 8i &= z^4 + z^2 - 8iz^2 - 8i \\ &= z^2(z^2+1) - 8i(z^2+1) \\ &= (z^2-8i)(z^2+1) \end{aligned}$$

$$z^2 = 8i \implies \boxed{z = \pm\sqrt{8i} \text{ with order } 1}$$

$$z^2 = -1 \implies \boxed{z = \pm i \text{ with order } 1}$$

Problem 16.2.9

Determine the location and order of the zeros.

$$\sin 2z \cos 2z$$

Solution

$$\sin 2z \cos 2z = \frac{1}{2} \sin 4z = 0 \implies \boxed{z = \frac{n\pi}{4}, \quad n \in \mathbb{Z}}$$

$$f'(z) = 2\cos 4z \qquad f'\left(\frac{n\pi}{4}\right) = 2(-1)^n \neq 0$$

Therefore the zeros are all simple zeros

Problem 16.2.13

Determine the location of the singularities, including those at infinity. For poles also state the order. Give reasons.

$$\frac{1}{(z+2i)^2} - \frac{z}{z-i} + \frac{z+1}{(z-i)^2}$$

Solution

Poles with order 2:

$$(z+2i)^2 \implies \boxed{z=-2i}$$

$$(z-i)^2 \implies \boxed{z=i}$$

Looking at ∞,

$$\begin{aligned} f(z) = f(1/w) &= \frac{1}{\left(\frac{1}{w}+2i\right)^2} - \frac{\frac{1}{w}}{\frac{1}{w}-i} + \frac{\frac{1}{w}+1}{\left(\frac{1}{w}-i\right)^2} \\ &= \frac{w^2}{(1+2iw)^2} - \frac{1}{1-iw} + \frac{w(1+w)}{(1-iw)^2} \end{aligned}$$

As $z \to \infty$ or $w \to 0$,

$$f \to 0 - 1 + 0 = -1$$

Therefore the function is bounded at ∞.

Problem 16.2.15

Determine the location of the singularities, including those at infinity. For poles also state the order. Give reasons.

$$z \exp\left(\frac{1}{(z-1-i)^2}\right)$$

Solution

Essential singularity:

$$(z-1-i)^2 \implies \boxed{z = 1+i}$$

Looking at ∞,

$$\begin{aligned} f(z) = f(1/w) &= \frac{1}{w} \exp\left(\frac{1}{\left(\frac{1}{w}-1-i\right)^2}\right) \\ &= \frac{1}{w} \exp\left(\frac{w^2}{(1-w-iw)^2}\right) \end{aligned}$$

As $z \to \infty$ or $w \to 0$ there is a simple pole.

Problem 16.2.17

Determine the location of the singularities, including those at infinity. For poles also state the order. Give reasons.

$$\cot^4 z$$

Solution

Poles with order 4:

$$\cot^4 z = \frac{\cos^4 z}{\sin^4 z} \implies \boxed{z = n\pi \quad n \in \mathbb{Z}}$$

Essential singularity at ∞

Problem 16.2.19

Determine the location of the singularities, including those at infinity. For poles also state the order. Give reasons.

$$\frac{1}{e^z - e^{2z}}$$

Solution

$$e^z - e^{2z} = 0 \implies (1 - e^z) = 0 \implies \boxed{z = 2n\pi i, \quad n \in \mathbb{Z}}$$

$$f'(z) = e^z - 2e^{2z} \qquad f'(2n\pi i) = -1$$

Therefore these are simple poles.

Essential singularity at ∞.

Problem 16.2.21

Determine the location of the singularities, including those at infinity. For poles also state the order. Give reasons.

$$\frac{e^{\frac{1}{z-1}}}{e^z - 1}$$

Solution

Essential singularity:

$$z - 1 = 0 \implies z = 1$$

Simple pole:

$$e^z - 1 = 0 \implies z = 2n\pi i, \quad n \in \mathbb{Z}$$

Problem 16.3.3

Find all the singularities in the finite plane and the corresponding residues.

$$\frac{\sin 2z}{z^6}$$

Solution

Problem 16.3.5

Find all the singularities in the finite plane and the corresponding residues.

$$\frac{8}{1+z^2}$$

Solution

Problem 16.3.7

Find all the singularities in the finite plane and the corresponding residues.

$$\cot \pi z$$

Solution

Problem 16.3.9

Find all the singularities in the finite plane and the corresponding residues.

$$\frac{1}{1-e^{z}}$$

Solution

Problem 16.3.11

Find all the singularities in the finite plane and the corresponding residues.

$$\frac{e^z}{(z - \pi i)^3}$$

Solution